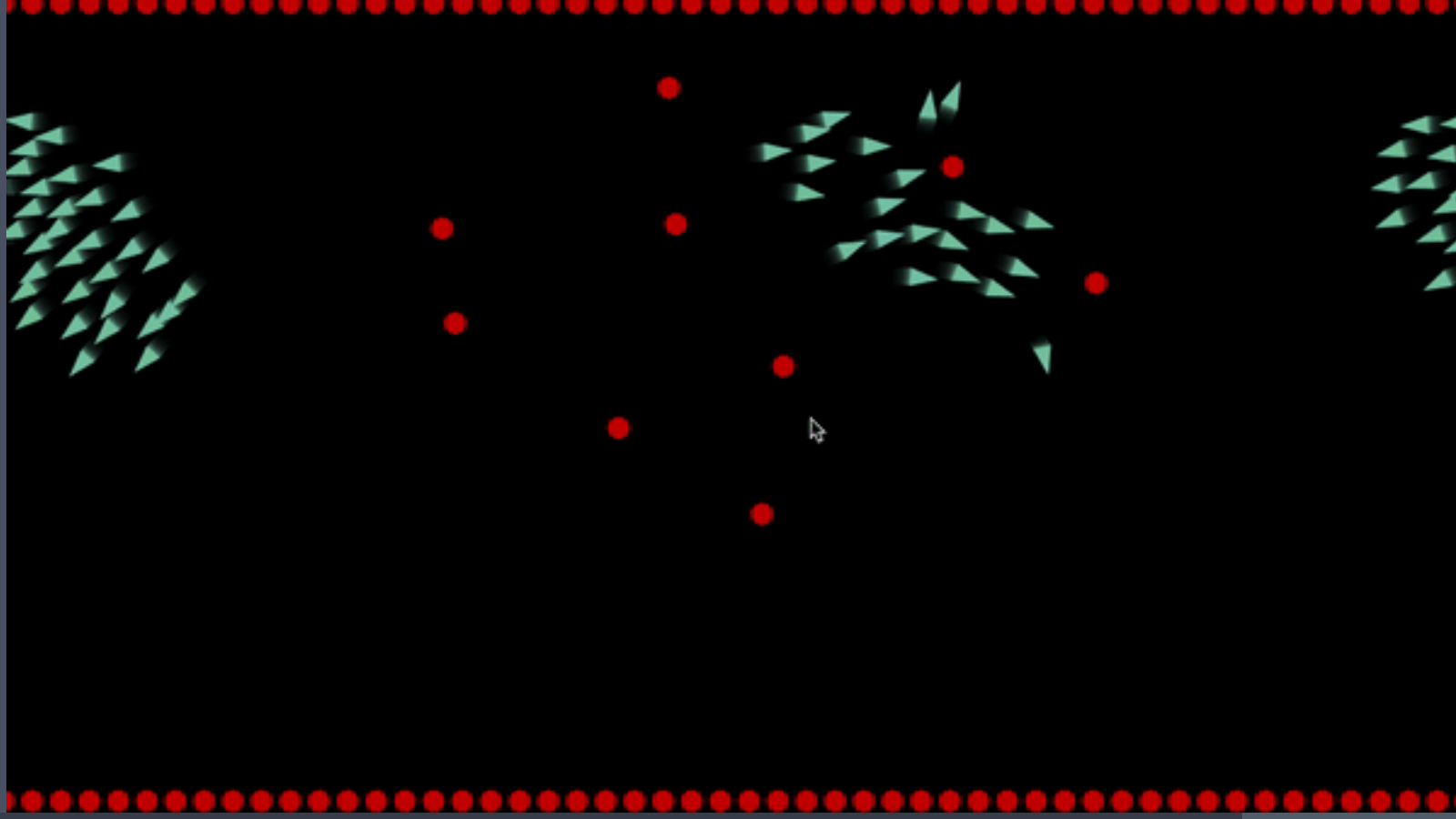




SENSS Specialist Training: Computational Simulations

20 May 2026
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Plan of Day 2

1. Building agent-based models
2. Application with AI (turn pseudo-code to computer code)
3. Complex adaptive systems and agent-based models
4. Exploration of further models

Computational Simulations for Theory Development

We have raised two points:

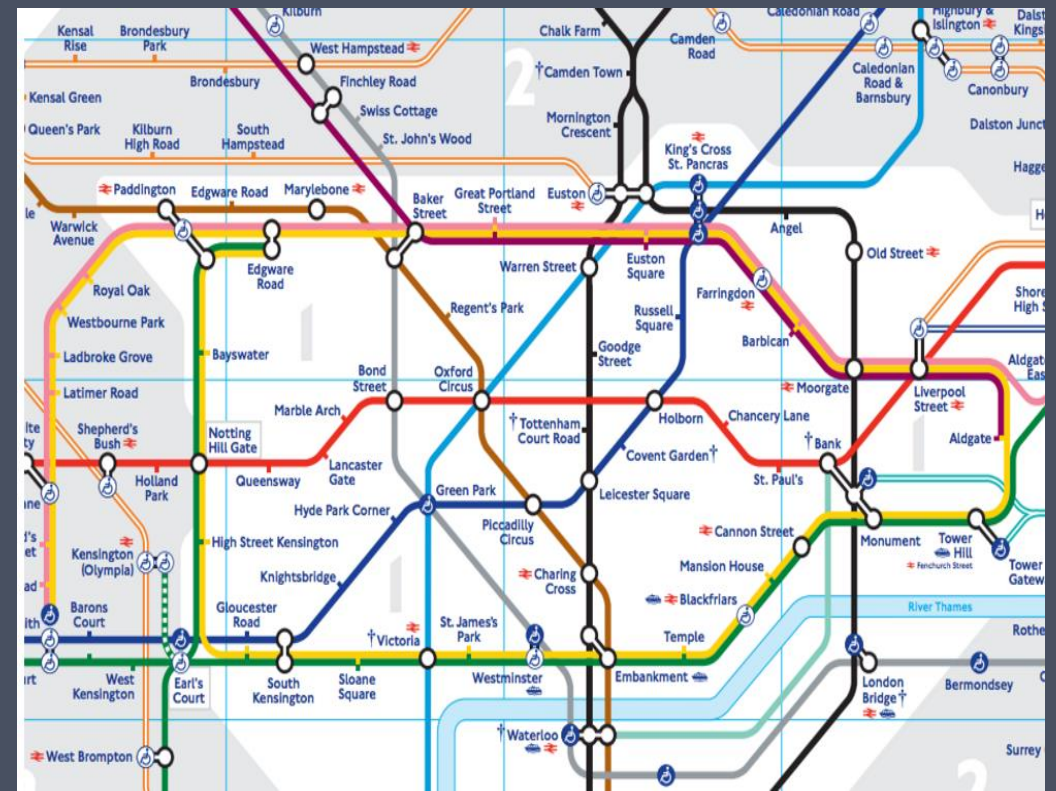
1. Social systems are complex systems
2. Computational simulation models aim to represent the complexity inherent in social systems

Good simulation models capture interactions that characterise complex social systems.

What is a model?

- A *model* is a *useful* simplification
- *useful* → we have a goal
- *goal* → *research question*
- Our goal is to understand a system (or parts of it)
 - *referent system*: the thing we are trying to understand

The goal of theory is to make the world understandable by finding the right set of simplifications.



Three Ways of Theoretical Modelling

1. Traditional theories: long, verbal descriptions

- Verbal models (Smith; Marx; many IR theories)
- High flexibility but low precision
- **Representation**: flexible (high-level of detail is possible but brings trade-offs)

2. Formal (mathematical models): commonly used in Economics

- High precision but low flexibility
- Simplified to a large extent: bringing trade-offs with applicability and **representation**
 - *For economists, the real world is often a special case*
- Not very good at dealing with **organised complexity**
- Top-down **abstraction**: high-level rules on the system (e.g., macro-economic models)

3. Computational models

- Middle ground: quite flexible and highly precise (code is formalisation)
- Representation: flexible
- Bottom-up: abstractions over the lower-level individual entities that make up the system.

Some terminology: abstraction and representation

Abstraction: the process of selecting a set of features from a referent system for modelling purposes

- conceptual curation: deciding what is essential and relevant
- simplification of the real-world system
- *abstraction produces a simplified conceptual representation of the system: **conceptual model***
- all theories are organised and systematic set of abstractions

Representation problem: finding a good way to represent the key elements of a real-world system in a model

- we may have a good idea which features of the real world are relevant
- but how to represent them in a model? *representation of abstractions so that a computer can process*
- necessary for moving from a conceptual model to a computational model
- *Much of the creativity in modelling lies in finding a representation that captures the core mechanisms without unnecessary detail*

Conceptual model and key abstractions in Schelling's segregation study

Conceptual model specifies the key elements of the system:

- the entities (agents)
- their attributes
- their rules of behaviour
- their interactions

Agents: households

Attribute: group identity (red / blue)

Preference rule: tolerance threshold

Interaction: neighbourhood composition

Key abstractions and their representation in Schelling's segregation model:

- Households represented by agents living in a grid-cell
- Race: agents' colour (red or blue)
- Agents only observe their local neighbours
 - only one type of information is received and processed by agents
- A single preference: don't want to be a minority below a tolerance threshold

Let's revisit the Research Question

- First-step in every scientific inquiry, including computational simulations

Every social simulation is intended to address one or more research questions about a referent system

- We often have multiple, nested research questions:
 - A main, overarching and extensive research question: broad phenomenon we want to understand
 - Specific and precise research questions nested in the main question
 - Models aim to understand the main question through addressing the nested, precise questions

Let's revisit the Research Question

Schelling's segregation model: what are the research question(s)?

- Why is racial segregation in cities so persistent?
 - Can benign individual preferences not to be too small of a minority contribute to racial segregation?
 - Explores a precise mechanism
- Research questions guide our conceptual model
 - abstractions we make are directly related to the research questions
 - what should be the scope of our abstractions → research question motivates this
 - Research questions are both the starting point and the anchor of scientific inquiry

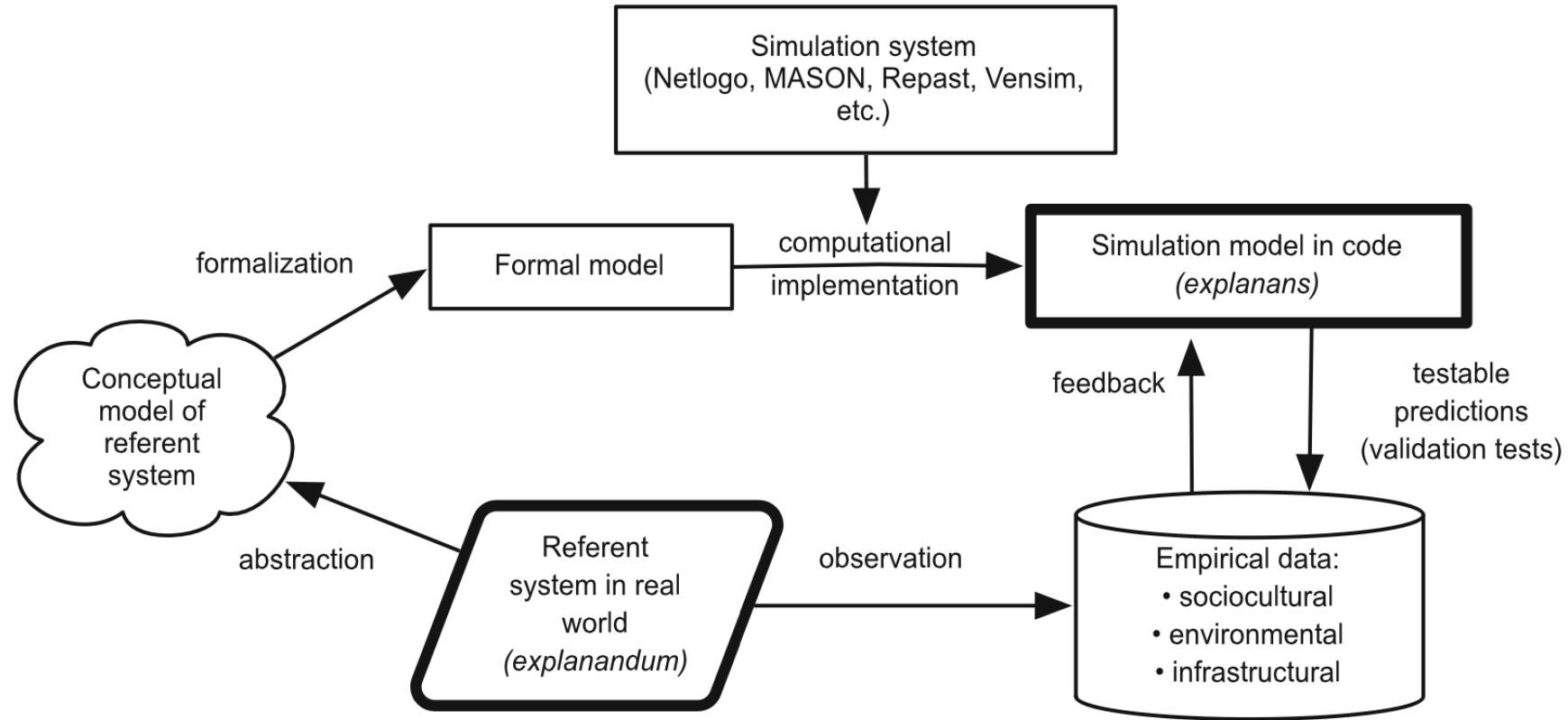


Fig. 8.1 *Basic terminology and general methodology of social simulation.* Social simulation methodology is an iterative process that begins with a referent system (*explanandum*) in the real world. Abstraction, formalization, programming, and appropriate data are used to develop a viable simulation model (*explanans*). This general process is independent of the specific kind of simulation model

Next: Formalization of the code

- Conceptual model → simulation model through the code
- **Formalization**: translating the conceptual model into computational code
- The model is expressed as precise and explicit rules

Pseudo-code

1. Create $n \times n$ grid
2. Create $n^2 \times \frac{90}{100}$ agents (10% is empty cells)
3. Each agent has a one attribute: colour $\in \{red, blue\}$
 - 40% red, 40% blue and 10% empty cells
4. Randomly assign each agent to a cell
5. Each cell is in one of the states: empty, red, blue
6. Agent decision rule (this is where agents process information and take action):
 - Each agent observes neighbouring cells
 - Compute the share of neighbours with the same colour
 - if share < tolerance threshold $t \rightarrow$ move to random empty cell
 - if share $\geq$ tolerance threshold $t \rightarrow$ stay where you are
7. Simulation loop: repeat the previous step until no agent wants to move (or max # of rounds)

Formalization of the code

- The choice of programming language is secondary
- C; Python; R; NetLogo; (game of life)
- Writing the code was the main barrier for researchers
 - large teams developed agent-based simulations
 - usually involved a software engineer / computer scientist
- This barrier has been rapidly decreasing
 - Learning programming is much easier
 - AI tools are extremely useful for learning and building *with*
- Yet, AI brings dangers: be careful

Tips for Using AI: Work with AI

- We still need to have a very strong command of programming
- *Build with AI: do not simply outsource the task to the AI*
- Each step needs to be clear, precise, transparent and reproducible
- Recreate each step and component (preferably manually)
- Back-and-forth with AI: discuss each step in detail
 - Ask AI to explain what it did in each step and why
- AI rarely understands or catches its own mistakes:
 - use multiple AI models (GPT, Claude, Gemini)
- Danger of black-box: AI creates something for you which you don't fully understand
 - This may render your simulation useless, or even worse, grossly misleading

Model Verification: Internal Validity

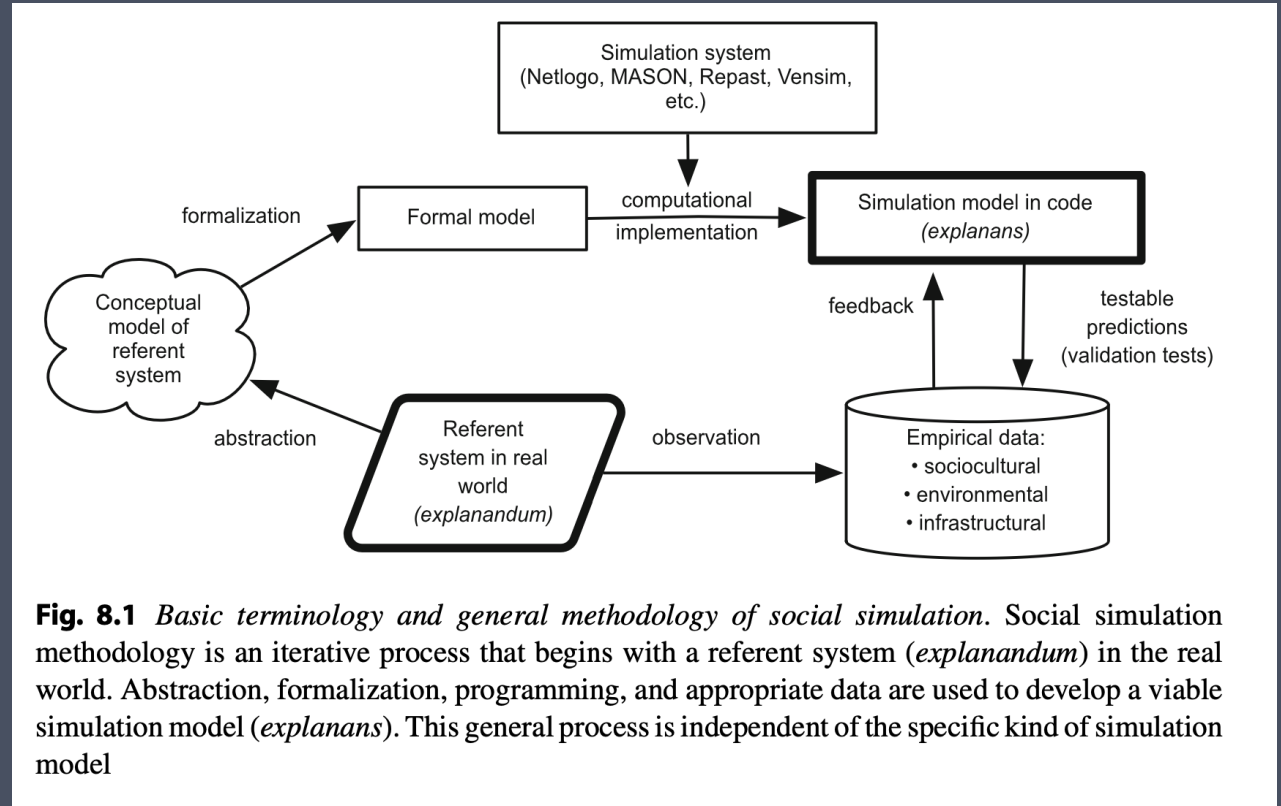
- We coded the model, but is it working the way it is intended?
- *Verification*: checking whether the simulation model correctly implements the conceptual model
- Ensuring the code does what the model specifies
- Identifying bugs or logical errors
- Testing whether rules behave as intended
- Closely verify the model
 - plug in scenarios
 - step by step follow the model (or a particular agent) closely
- Tips about using AI are highly pertinent for verification

Model Validation: External Validity

- Let's say the model is running as we intended: we have high internal validity
- But does the model produce patterns or outcomes we see in the referent system?
 - Does it represent the real-world system?
 - This is related to the *external validity*
- **Validation:** *assessing whether the model captures relevant features or patterns of the referent system.*
- Comparing *simulation outcomes* with *empirical observations*
- Checking whether the model reproduces known patterns or regularities
- **Schelling's model:** does the model generate segregated neighbourhoods similar to those observed in real cities?

Model Validation: Calibration

- Some models may need parameter calibration
- What are some plausible parameter values?
- We can rely on empirical data
- **Calibration**: estimating and adjusting parameter values so that they fall within plausible ranges and improve the model's fit to observed patterns
- Validation often involves calibrating parameter values using empirical data



Calibration continued

- **Schelling's model:** what is the real-world tolerance threshold τ in the city we are studying?
- There can be cultural differences across time and space; leading to different τ :
 - Harrison, Arkansas vs Seattle, Washington
 - London 1960 vs London 2026
- Maybe we can use survey data: questions related to attitudes towards neighbours



Next: Formalization of the Code

- Let's have an applied example
- Recall your task from yesterday.
- Give it to an AI model (Google Colab, Gemini)